

# Uranus

## Overview

Uranus is the seventh planet from the Sun, orbiting at an average distance of 2.87 billion km and taking 84 Earth years to complete one orbit. With a diameter of 50,724 km (about four times Earth's), it is the third-largest planet in the Solar System by diameter and the fourth-largest by mass. Uranus is classified as an ice giant rather than a gas giant, because while its outer layers are hydrogen and helium gas, its interior is thought to be dominated by a slushy mixture of water, methane, and ammonia ices under extreme pressure, surrounding a small rocky core. Uranus was the first planet discovered with a telescope, by William Herschel on 13 March 1781; all planets closer to the Sun had been known since antiquity because they are visible to the naked eye.

## Extreme Axial Tilt and Seasons

The most remarkable characteristic of Uranus is its extreme axial tilt of  $97.77^\circ$ , meaning it rotates almost on its side relative to its orbital plane around the Sun. Most planets rotate with their poles roughly perpendicular to the plane of the Solar System, but Uranus's poles point nearly directly toward and away from the Sun at different points in its orbit. The leading explanation is a massive collision with an Earth-sized protoplanet early in the Solar System's history that knocked Uranus onto its side. As a consequence of this tilt, each pole of Uranus experiences approximately 42 years of continuous sunlight followed by 42 years of total darkness during its 84-year orbit. Counterintuitively, despite the poles receiving more direct sunlight than the equator for much of the orbit, the equator is actually warmer due to complex atmospheric dynamics not yet fully understood. During Voyager 2's flyby in 1986, the south pole was pointed almost directly at the Sun, giving scientists a unique view of a near-pole-on planet.

## Atmosphere and Interior

Uranus's upper atmosphere is composed of about 83 % hydrogen, 15 % helium, and 2 % methane. Methane in the upper atmosphere absorbs red wavelengths of sunlight and reflects blue-green light, giving Uranus its distinctive pale cyan colour. Unlike Jupiter, Saturn, and Neptune, Uranus emits almost no detectable internal heat; it radiates roughly the same amount of energy it receives from the Sun, which is unusual and not fully explained. Because of this lack of internal heating, Uranus has the coldest planetary atmosphere in the Solar System, with temperatures in the upper troposphere reaching  $-224^\circ\text{C}$  — colder even than Neptune, despite Neptune being farther from the Sun. Scientists hypothesise that some insulating layer deep in Uranus's interior may be trapping its primordial heat, preventing it from escaping. Under extreme pressures deep within the ice mantle, water, ammonia, and methane may break down into ionic forms, and some models predict that pure carbon could crystallise into diamond at depths of thousands of kilometres.

## Rings and Moons

Uranus has 13 known rings, which are narrow, dark, and composed of relatively large chunks of material rather than fine dust. They were discovered in 1977 when astronomers observed Uranus passing in front of a star and noticed the star's light flickered both before and after the planet crossed in front of it — indicating ring material on both sides. The rings are thought to be geologically young and may originate from the fragmentation of small moons. Uranus has 28 known moons, all named after characters from the works of William Shakespeare and the poet Alexander Pope — a tradition unique in the Solar System. The five major moons, in order of increasing distance, are Miranda, Ariel, Umbriel, Titania, and Oberon. Miranda is the most geologically diverse, featuring enormous cliffs up to 20 km tall — the tallest known cliffs in the Solar System — and a patchwork of dramatically different terrain types, suggesting a violent geological history.